Gyires's permanent inequality for nonnegative circulant matrices of order five

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Abstract

Gyires's conjecture gives a lower bound for a ratio of permanents of doubly stochastic matrices. The result holds for every nonnegative circulant matrix of order five, including matrices with zero entries. Every nonnegative circulant C of order five satisfies the stronger inequality $24\,\mathrm{per}(CC^{\mathsf{T}}) \le 625\,\mathrm{per}(C)^2$. Equality holds exactly when all entries of C are equal. A sharp bound for the difference also holds. Let $z_1, \dots, z_4$ be the gaps between successive sorted weights of C. Then $625\,\mathrm{per}(C)^2 - 24\,\mathrm{per}(CC^{\mathsf{T}}) \ge 484 \sum_{j=1}^{4} z_j^{10}$, and the constant 484 cannot be increased. An affine change of indices reduces the proof to six orderings of the weights. In each ordering, the difference is a polynomial with nonnegative integer coefficients. The principal inequalities, equality cases, and sharp constant have complete Lean 4 proofs. The result concerns this restricted matrix family. It does not settle the full conjecture.

Authorship and review. Dev D. Goyal directed the research and accepts responsibility for the paper. The original project records that ChatGPT (OpenAI), using the GPT-6 model with Pro reasoning, found the proof and wrote the first formalization and draft. It records that Claude Opus 5.5, used with Claude Code, revised the paper, ported the proof, and performed additional checks. Those checks were not independent of the files that Claude itself wrote. This revision preserves that historical disclosure and separates it from the current audit. The paper has not received independent expert or journal peer review. The final section gives the contribution record and review limits.

1 Introduction

1.1 The conjecture

A real square matrix is *doubly stochastic* if its entries are nonnegative and all its row sums and column sums equal 1. Write Ω_n for the set of doubly stochastic $n \times n$ matrices, $J_n \in \Omega_n$ for the matrix all of whose entries equal $1/n$, and

$$\mathrm{per}\, A = \sum_{\sigma \in S_n} \prod_{i=1}^{n} a_{i\sigma(i)}$$

for the permanent of $A = (a_{ij})$. By van der Waerden's conjecture, proved independently by Gyires [9], Egorychev [5] and Falikman [6], $\mathrm{per}\, A \ge n!/n^n$ for every $A \in \Omega_n$, with equality only at J_n. B. Gyires [8] proposed a related lower bound for the square of the permanent. It is Conjecture 20 in the survey of open problems on permanents by Cheon and Wanless [3], in the following form, where A^* denotes the conjugate transpose.

Conjecture 1.1 (Gyires; Cheon and Wanless [3, Conjecture 20]). *For every $A \in \Omega_n$,*

$$\frac{4\,\mathrm{per}(A)^2}{\mathrm{per}(AA^*) + \mathrm{per}(A^*A) + 2\,\mathrm{per}(A^2)} \;\geq\; \frac{n!}{n^n},$$

with equality if and only if $A = J_n$.

For real matrices $A^* = A^{\mathsf{T}}$, the transpose, and this is the form used below. For $A \in \Omega_n$ the matrices AA^{T}, $A^{\mathsf{T}}A$ and A^2 are doubly stochastic, so by van der Waerden's theorem each of their permanents is at least $n!/n^n$, and the denominator is positive. The conjecture is therefore equivalent to the polynomial inequality

$$\frac{n!}{n^n}\bigl(\mathrm{per}(AA^{\mathsf{T}}) + \mathrm{per}(A^{\mathsf{T}}A) + 2\,\mathrm{per}(A^2)\bigr) \;\leq\; 4\,\mathrm{per}(A)^2 \qquad (A \in \Omega_n), \tag{1}$$

with equality only at J_n. Call (1) the *homogeneous form* of Gyires's inequality: both sides are homogeneous of degree $2n$ in the entries of A, so it makes sense for every square matrix.

1.2 What this paper proves

A matrix C of order five is *circulant* if its entries depend only on the difference of the indices modulo five: $c_{ij} = w_{j-i}$ for a vector $w = (w_0, \ldots, w_4) \in \mathbb{R}^5$, the indices being read in $\mathbb{Z}/5\mathbb{Z}$. Call w, the first row of C, its vector of *weights*, and write $C = C(w)$. A circulant is doubly stochastic exactly when $w \geq 0$ and $w_0 + \cdots + w_4 = 1$, so the doubly stochastic circulants of order five are the convex combinations $\sum_k w_k Q^k$ of the powers of the cyclic permutation matrix $Q = C(0,1,0,0,0)$. They form a four-dimensional simplex inside the sixteen-dimensional polytope Ω_5; this simplex contains matrices with zero entries, among them its vertices Q^k. For $n = 5$ the constant of the conjecture is $5!/5^5 = 24/625$.

Main results. *Let $C = C(w)$ be a circulant of order five with nonnegative weights.*

(a) (Theorem 3.1, the strong inequality.) $24\,\mathrm{per}(CC^{\mathsf{T}}) \leq 625\,\mathrm{per}(C)^2$, with equality if and only if all five weights are equal.

(b) (Theorem 3.2, the margin.) If $w_{\pi(0)} \leq w_{\pi(1)} \leq \cdots \leq w_{\pi(4)}$ for a permutation π and $z_j = w_{\pi(j)} - w_{\pi(j-1)}$ for $1 \leq j \leq 4$, then $625\,\mathrm{per}(C)^2 - 24\,\mathrm{per}(CC^{\mathsf{T}}) \geq 484\,(z_1^{10} + z_2^{10} + z_3^{10} + z_4^{10})$, and no constant larger than 484 is possible.

(c) (Corollary 3.3.) The homogeneous form (1) of Gyires's inequality holds for C, with equality if and only if all five weights are equal. In particular Conjecture 1.1 holds for every doubly stochastic circulant of order five, including its equality case.

The strong inequality (a) implies Gyires's inequality for C: for every real circulant C one has $CC^{\mathsf{T}} = C^{\mathsf{T}}C$ and $\mathrm{per}(C^2) \leq \mathrm{per}(CC^{\mathsf{T}})$ (Corollary 2.4), so the left-hand side of (1) is at most $\frac{24}{625} \cdot 4\,\mathrm{per}(CC^{\mathsf{T}})$, which by (a) is at most $4\,\mathrm{per}(C)^2$.

1.3 Prior work and scope

For $n = 2$, write $A = \left(\begin{smallmatrix} a & 1-a \\ 1-a & a \end{smallmatrix}\right)$ with $0 \leq a \leq 1$. Put $s = a^2 + (1-a)^2$ and $t = 2a(1-a)$. Then $\mathrm{per}\, A = s$ and each permanent in the denominator is $s^2 + t^2$. The conjecture reduces to $s \geq t$, or $(2a-1)^2 \geq 0$. Equality holds only at J_2. Chang [2] proved the order-three inequality. His sufficient condition in every order requires each entry to be at least $(n-2)/(n-1)^2$. For order-five circulants, this condition covers weights in $[3/16, 1/4]$. It excludes every zero weight.

Ozols's public development [12] gives the full order-four inequality and equality case. It also gives $3\,\mathrm{per}(AA^{\mathsf{T}}) \le 32\,\mathrm{per}(A)^2$. Its certificate uses `native_decide`, which is outside the proof rule used for the present development. It remains relevant prior work.

Lavi's current catalogue [10] lists the general conjecture as open. It announces boundary and neighborhood results, certain symmetric families, all-order mixtures of two permutations, and mixtures of three permutations in order five. These statements have narrower or different hypotheses from the theorem below. Their omission of a result is not evidence of novelty.

Minc [11] already gave exact permanent formulas for three-term circulants. The present contribution is the order-five Gram inequality, its equality case, and the sharp sorted-gap constant. This is a revision of the existing result in [7], not a new theorem from the current audit. The prior-work comparison remains qualified by the source-access limits stated at the end.

1.4 Motivation

The general conjecture remains open in order five and above. Circulants reduce the four matrix permanents to polynomials in five weights. They also satisfy $CC^{\mathsf{T}} = C^{\mathsf{T}}C$. These properties make them a useful restricted family. Their permanents have a long history; Minc [11] treated three-term circulants. The present family has four independent parameters after normalization. A finite sample of matrices therefore cannot prove the theorem. The sorted-gap coordinates give an exact certificate over the whole family, including its boundary.

1.5 How to read this paper

Section 2 collects the identities for circulants and a comparison lemma for permanents. Section 3 states the results, and Section 4 proves them. Section 5 describes the computer checks and the formal proof, and Section 6 discusses the scope of the results.

2 Background

2.1 Notation

Indices of rows, columns and weights are taken in $\mathbb{Z}/5\mathbb{Z} = \{0,1,2,3,4\}$. For $w \in \mathbb{R}^5$, write $C(w)$ for the circulant with entries $C(w)_{ij} = w_{j-i}$, and

$$P(w) = \operatorname{per} C(w), \qquad g(w)_k = \sum_{i=0}^{4} w_i w_{i+k}, \qquad h(w)_k = \sum_{i=0}^{4} w_i w_{k-i} \qquad (k \in \mathbb{Z}/5\mathbb{Z}),$$

so that $g(w)$ is the circular autocorrelation of w and $h(w)$ its circular self-convolution. Note that $g(w)_{-k} = g(w)_k$. The *defects*, D_s for the strong inequality and D_G for Gyires's inequality, are the homogeneous polynomials of degree ten

$$D_s(w) = 625\,P(w)^2 - 24\,P(g(w)), \qquad D_G(w) = 625\,P(w)^2 - 12\big(P(g(w)) + P(h(w))\big). \quad (2)$$

Lemma 2.1. *For every $w \in \mathbb{R}^5$, $C(w)C(w)^{\mathsf{T}} = C(w)^{\mathsf{T}}C(w) = C(g(w))$ and $C(w)^2 = C(h(w))$.*

Proof. Write $C = C(w)$. Substituting $m = k - j$, respectively $m = i - k$ and $m = k - i$,

$$(CC^{\mathsf{T}})_{ij} = \sum_k w_{k-i} w_{k-j} = \sum_m w_m w_{m+j-i} = g_{j-i},$$
$$(C^{\mathsf{T}}C)_{ij} = \sum_k w_{i-k} w_{j-k} = \sum_m w_m w_{m+j-i} = g_{j-i},$$

and $(C^2)_{ij} = \sum_k w_{k-i} w_{j-k} = \sum_m w_m w_{(j-i)-m} = h_{j-i}$. □

Lemma 2.2. *For every $w = (w_0, \dots, w_4) \in \mathbb{R}^5$, writing $w = (a, b, c, d, e)$,*

$$\begin{aligned} P(w) = {} & a^5 + b^5 + c^5 + d^5 + e^5 + 15\,abcde \\ & + 5\big(a^3be + a^3cd + a^2b^2d + a^2bc^2 + a^2ce^2 + a^2d^2e + ab^3c + ab^2e^2 + abd^3 + ac^3e \\ & \quad + ac^2d^2 + ade^3 + b^3de + b^2c^2e + b^2cd^2 + bc^3d + bce^3 + bd^2e^2 + c^2de^2 + cd^3e\big). \end{aligned}$$

Proof. By definition $P(w) = \sum_{\sigma \in S_5} \prod_i w_{\sigma(i)-i}$, a sum of 120 monomials of degree five; collecting them gives the 26 terms displayed. For instance, the five translations $\sigma(i) = i + k$ give w_k^5, and the monomial *abcde* arises from the 15 permutations σ for which $i \mapsto \sigma(i) - i$ is a bijection of $\mathbb{Z}/5\mathbb{Z}$. The coefficients add up to $120 = 5!$. □

The following comparison is used to pass from the stronger inequality to Gyires's inequality.

Lemma 2.3. *For real $n \times n$ matrices M and N, $2\,\mathrm{per}(MN^{\mathsf{T}}) \le \mathrm{per}(MM^{\mathsf{T}}) + \mathrm{per}(NN^{\mathsf{T}})$.*

Proof. Let $x_1, \dots, x_n$ be indeterminates and define coefficients μ_α and ν_α, indexed by $\alpha \in \mathbb{N}^n$ with $\alpha_1 + \cdots + \alpha_n = n$, by

$$\prod_{i=1}^n \Big(\sum_{k=1}^n M_{ik} x_k\Big) = \sum_\alpha \mu_\alpha x^\alpha, \qquad \prod_{i=1}^n \Big(\sum_{k=1}^n N_{ik} x_k\Big) = \sum_\alpha \nu_\alpha x^\alpha.$$

Thus $\mu_\alpha = \sum_\varphi \prod_i M_{i\varphi(i)}$, the sum over the maps $\varphi\colon \{1, \dots, n\} \to \{1, \dots, n\}$ with occupancy α, that is, with $|\varphi^{-1}(k)| = \alpha_k$ for every k; similarly for ν_α. Expanding,

$$\mathrm{per}(MN^{\mathsf{T}}) = \sum_{\sigma \in S_n} \prod_i \sum_k M_{ik} N_{\sigma(i)k} = \sum_{\sigma \in S_n} \sum_\varphi \prod_i M_{i\varphi(i)} \prod_j N_{j,\varphi(\sigma^{-1}(j))}.$$

For fixed φ with occupancy α, the map $\sigma \mapsto \varphi \circ \sigma^{-1}$ takes every map of occupancy α exactly $\alpha! = \prod_k \alpha_k!$ times, and no other map. Hence

$$\mathrm{per}(MN^{\mathsf{T}}) = \sum_\alpha \alpha!\, \mu_\alpha \nu_\alpha. \tag{3}$$

Applying (3) also to (M, M) and (N, N), the claim follows from $2\mu_\alpha\nu_\alpha \le \mu_\alpha^2 + \nu_\alpha^2$ and $\alpha! > 0$. □

Corollary 2.4. *For every $w \in \mathbb{R}^5$, $P(h(w)) \le P(g(w))$. Consequently $D_G(w) \ge D_s(w)$.*

Proof. Take $M = C(w)$ and $N = C(w)^{\mathsf{T}}$ in Lemma 2.3. By Lemma 2.1, $MN^{\mathsf{T}} = C(h(w))$ and $MM^{\mathsf{T}} = NN^{\mathsf{T}} = C(g(w))$, so $2P(h(w)) \le 2P(g(w))$. Finally $D_G - D_s = 12\big(P(g) - P(h)\big)$. □

3 Results

Theorem 3.1. *Let $w \in \mathbb{R}^5$ with $w_i \geq 0$ for all i, and let $C = C(w)$. Then*

$$24\,\mathrm{per}(CC^{\mathsf{T}}) \leq 625\,\mathrm{per}(C)^2, \qquad \textit{that is,} \qquad \frac{5!}{5^5}\,\mathrm{per}(CC^{\mathsf{T}}) \leq \mathrm{per}(C)^2.$$

Equality holds if and only if $w_0 = w_1 = w_2 = w_3 = w_4$.

Theorem 3.2. *Let $w \in \mathbb{R}^5$ with $w_i \geq 0$ for all i, let $C = C(w)$, let π be a permutation of $\mathbb{Z}/5\mathbb{Z}$ with $w_{\pi(0)} \leq w_{\pi(1)} \leq \cdots \leq w_{\pi(4)}$, and put $z_j = w_{\pi(j)} - w_{\pi(j-1)}$ for $1 \leq j \leq 4$. Then*

$$D_s(w) = 625\,\mathrm{per}(C)^2 - 24\,\mathrm{per}(CC^{\mathsf{T}}) \;\geq\; 484\,(z_1^{10} + z_2^{10} + z_3^{10} + z_4^{10}).$$

Equality holds for $w = (1, 1, 0, 0, 0)$, for which $z_1^{10} + \cdots + z_4^{10} = 1$ and $D_s(w) = 484$. Consequently, if a real constant K satisfies $D_s(w) \geq K\,(z_1^{10} + \cdots + z_4^{10})$ for all such w and π, then $K \leq 484$.

Corollary 3.3. *Let $w \in \mathbb{R}^5$ with $w_i \geq 0$ for all i, and let $C = C(w)$. Then*

$$24\big(\mathrm{per}(CC^{\mathsf{T}}) + \mathrm{per}(C^{\mathsf{T}}C) + 2\,\mathrm{per}(C^2)\big) \leq 2500\,\mathrm{per}(C)^2,$$

with equality if and only if all five weights are equal. If moreover $w_0 + \cdots + w_4 = 1$, then $C \in \Omega_5$, the denominator below is positive, and

$$\frac{4\,\mathrm{per}(C)^2}{\mathrm{per}(CC^{\mathsf{T}}) + \mathrm{per}(C^{\mathsf{T}}C) + 2\,\mathrm{per}(C^2)} \;\geq\; \frac{24}{625} = \frac{5!}{5^5},$$

with equality if and only if $C = J_5$. Thus Conjecture 1.1 holds for every doubly stochastic circulant matrix of order five.

4 Proofs

The idea in brief. By Corollary 2.4 it suffices to bound D_s. The defect is invariant under the affine changes of indices $i \mapsto ai + b$ of $\mathbb{Z}/5\mathbb{Z}$, and these reduce the 120 possible orderings of the weights to six. In each of the six chambers, the defect written in the smallest weight and the gaps between consecutive sorted weights is a polynomial with nonnegative integer coefficients, which proves the inequality, the margin and the equality case at once.

4.1 Affine symmetry

For $a \in (\mathbb{Z}/5\mathbb{Z})^\times$ and $b \in \mathbb{Z}/5\mathbb{Z}$ let $(T_{a,b}w)_i = w_{ai+b}$; the twenty maps $i \mapsto ai + b$ are the affine bijections of $\mathbb{Z}/5\mathbb{Z}$.

Lemma 4.1. *For all $w \in \mathbb{R}^5$, a and b: $P(T_{a,b}w) = P(w)$, $g(T_{a,b}w)_k = g(w)_{ak}$ and $h(T_{a,b}w)_k = h(w)_{ak+2b}$. Consequently $D_s(T_{a,b}w) = D_s(w)$ and $D_G(T_{a,b}w) = D_G(w)$.*

Proof. The identity $C(T_{a,b}w)_{ij} = w_{a(j-i)+b} = C(w)_{ai,\,aj+b}$ shows that $C(T_{a,b}w)$ arises from $C(w)$ by permuting the rows by $i \mapsto ai$ and the columns by $j \mapsto aj + b$; permanents are invariant under such permutations. Substituting $m = ai + b$,

$$g(T_{a,b}w)_k = \sum_i w_{ai+b}\,w_{a(i+k)+b} = \sum_m w_m\,w_{m+ak},$$
$$h(T_{a,b}w)_k = \sum_i w_{ai+b}\,w_{a(k-i)+b} = \sum_m w_m\,w_{ak+2b-m}.$$

Hence $g(T_{a,b}w) = T_{a,0}\,g(w)$ and $h(T_{a,b}w) = T_{a,2b}\,h(w)$, and the first part gives $P(g(T_{a,b}w)) = P(g(w))$ and $P(h(T_{a,b}w)) = P(h(w))$. □

4.2 Six chambers

A *canonical ordering* is one of the six sequences

$$01234, \quad 01243, \quad 01324, \quad 01342, \quad 01423, \quad 01432, \tag{4}$$

that is, an ordering $\kappa = (\kappa(0), \dots, \kappa(4))$ of $\mathbb{Z}/5\mathbb{Z}$ with $\kappa(0) = 0$ and $\kappa(1) = 1$. For a canonical ordering κ and $z = (z_0, \dots, z_4) \in \mathbb{R}^5$ let $L_\kappa(z) \in \mathbb{R}^5$ be the vector with

$$L_\kappa(z)_{\kappa(j)} = z_0 + z_1 + \cdots + z_j \qquad (0 \le j \le 4).$$

If $z \ge 0$, the weights of $L_\kappa(z)$ are nondecreasing along κ, z_0 is the smallest weight and $z_1, \dots, z_4$ are the gaps between consecutive sorted weights. The *chamber* of κ is the set $\{L_\kappa(z) : z \ge 0\}$ of nonnegative weight vectors that are nondecreasing along κ.

Lemma 4.2. *Let π be a permutation of $\mathbb{Z}/5\mathbb{Z}$. There are $a \in (\mathbb{Z}/5\mathbb{Z})^\times$, $b \in \mathbb{Z}/5\mathbb{Z}$ and a canonical ordering κ with $\pi(j) = a\kappa(j) + b$ for every j, and the triple (a, b, κ) is unique. If $w \in \mathbb{R}^5$ satisfies $w_{\pi(0)} \le \cdots \le w_{\pi(4)}$ and $w_{\pi(0)} \ge 0$, then $T_{a,b}w = L_\kappa(z)$ with $z_0 = w_{\pi(0)}$ and $z_j = w_{\pi(j)} - w_{\pi(j-1)}$ $(1 \le j \le 4)$, and $z \ge 0$.*

Proof. Put $b = \pi(0)$ and $a = \pi(1) - \pi(0) \ne 0$, and $\kappa(j) = a^{-1}(\pi(j) - b)$. Then κ is an ordering of $\mathbb{Z}/5\mathbb{Z}$ with $\kappa(0) = 0$ and $\kappa(1) = 1$, hence canonical, and $\pi(j) = a\kappa(j) + b$; conversely the conditions $\kappa(0) = 0$ and $\kappa(1) = 1$ force $b = \pi(0)$ and $a = \pi(1) - \pi(0)$. For the second statement, $(T_{a,b}w)_{\kappa(j)} = w_{a\kappa(j)+b} = w_{\pi(j)} = z_0 + \cdots + z_j$, and $z \ge 0$ because the sorted weights are nondecreasing and the smallest is nonnegative. □

4.3 The chamber certificate

Lemma 4.3. *For each canonical ordering κ, the polynomial*

$$R_\kappa(z) = D_s(L_\kappa(z)) - 484\,(z_1^{10} + z_2^{10} + z_3^{10} + z_4^{10})$$

in $z = (z_0, \dots, z_4)$ has nonnegative integer coefficients. More precisely, $D_s(L_\kappa(z))$ is homogeneous of degree ten, so it has $\binom{14}{4} = 1001$ coefficients; 996 of them are positive, and the five coefficients of z_0^{10} and $z_0^9 z_j$ $(1 \le j \le 4)$ are zero. The coefficients of $z_1^{10}, z_2^{10}, z_3^{10}, z_4^{10}$ are 233 296, 44 209, 484 and 601, and every positive coefficient is at least 484.

Proof. This is a finite computation with integer polynomials. By Lemma 2.2, $P(w)$ is an explicit integer polynomial, and by definition so is $P(g(w))$; one substitutes $w = L_\kappa(z)$ into $625P(w)^2 - 24P(g(w))$, expands, and reads off the 1001 coefficients, for each of the six orderings (4). All coefficient claims of the lemma are obtained in this way. The statement about R_κ follows: the four pure powers z_j^{10}, $j \ge 1$, have coefficients at least 484, and every other coefficient is unchanged. □

The computation can be organized so that it is small enough to be checked by a proof assistant, and this is how it is done in the formal proof (Section 5). Let $u = L_\kappa(0, z_1, \dots, z_4)$, so that $L_\kappa(z) = z_0\mathbf{1} + u$, where $\mathbf{1} = (1, \dots, 1)$. Two identities make the dependence on z_0 explicit. First, $P(t\mathbf{1} + u) = \sum_{k=0}^{5} p_k(u)\, t^k$ for explicit polynomials p_k. Second, $g(t\mathbf{1} + u) = (5t^2 + 2tS)\mathbf{1} + g(u)$, where $S = u_0 + \cdots + u_4$, so that $P(g(t\mathbf{1}+u)) = \sum_{k=0}^{5} p_k(g(u))\,(5t^2 + 2tS)^k$. Hence $D_s(L_\kappa(z)) = \sum_{j=0}^{10} Q_j\, z_0^j$ with explicit polynomials Q_j in $z_1, \dots, z_4$, and each Q_j is expanded in powers of z_1 with coefficients that are polynomials in z_2, z_3, z_4; by Lemma 4.3, $Q_9 = Q_{10} = 0$. The nonnegativity of the coefficients of R_κ amounts to the statement that, after $484\,(z_1^{10} + \cdots + z_4^{10})$ is subtracted from Q_0, each of these polynomials in z_2, z_3, z_4 has nonnegative coefficients.

4.4 Proof of Theorems 3.2 and 3.1

Proof of Theorem 3.2. Let $w \geq 0$ and π be as in the theorem, and choose a, b, κ and z by Lemma 4.2. By Lemmas 4.1 and 4.3, since $z \geq 0$,

$$D_s(w) = D_s(T_{a,b}w) = D_s(L_\kappa(z)) = 484\,(z_1^{10} + \cdots + z_4^{10}) + R_\kappa(z) \;\geq\; 484\,(z_1^{10} + \cdots + z_4^{10}).$$

For the equality case, let $w = (1, 1, 0, 0, 0)$, sorted by $\pi = (2, 3, 4, 0, 1)$, with gaps $(z_1, z_2, z_3, z_4) = (0, 0, 1, 0)$. Here $g(w) = (2, 1, 0, 0, 1)$, and Lemma 2.2 gives $P(w) = 2$ and $P(g(w)) = 84$, so $D_s(w) = 625 \cdot 4 - 24 \cdot 84 = 484$. A constant K as in the theorem therefore satisfies $K \cdot 1 \leq 484$. □

Proof of Theorem 3.1. By Lemma 2.1, $\operatorname{per}(CC^{\mathsf{T}}) = P(g(w))$, so the inequality is $D_s(w) \geq 0$, which follows from Theorem 3.2 applied with any permutation π that sorts w. If $D_s(w) = 0$, Theorem 3.2 gives $z_1 = \cdots = z_4 = 0$, so all weights are equal. Conversely, if every weight equals c, then $P(w) = 120c^5$ (each of the 120 permutations contributes c^5) and $g(w) = (5c^2, \ldots, 5c^2)$, so $P(g(w)) = 120 \cdot 5^5 c^{10}$ and $625P(w)^2 = 9\,000\,000\,c^{10} = 24P(g(w))$. □

4.5 Proof of Corollary 3.3

Proof. By Lemma 2.1, $\operatorname{per}(CC^{\mathsf{T}}) + \operatorname{per}(C^{\mathsf{T}}C) + 2\operatorname{per}(C^2) = 2P(g(w)) + 2P(h(w))$, so the first inequality reads $4D_G(w) \geq 0$. By Corollary 2.4 and Theorem 3.1, $D_G(w) \geq D_s(w) \geq 0$. If $D_G(w) = 0$, then $D_s(w) = 0$, and all weights are equal by Theorem 3.1; conversely, for constant w one has $h(w) = g(w)$, hence $D_G(w) = D_s(w) = 0$.

Now let also $\sum_i w_i = 1$. Every row and every column of C contains each weight once, so $C \in \Omega_5$. Some weight is positive, so $g(w)_0 = \sum_i w_i^2 > 0$. The matrix $C(g(w))$ has nonnegative entries and diagonal entries $g(w)_0$, so $P(g(w)) \geq g(w)_0^5 > 0$, the identity permutation contributing $g(w)_0^5$ and the others nonnegative terms. Since also $P(h(w)) \geq 0$, the denominator $2P(g(w)) + 2P(h(w))$ is positive, and dividing the first inequality by 625 times the denominator gives the ratio inequality. Equality holds if and only if all weights are equal, that is, equal to $1/5$, which means $C = J_5$. □

5 Checking the results

The verification has two parts: exact polynomial computation and formal proof. The text distinguishes the original project records from checks performed for this revision.

Exact computations. The original project reports 72 checks with a PARI/GP script in exact integer and rational arithmetic. The present audit did not rerun that full PARI/GP script. It was written separately from the Lean files and does not use them, or the program that generated the certificate files; it is the file `independent_checks.gp` in the directory `verification` of the accompanying repository [7]. It computes every permanent from the definition, as a sum over all permutations, and compares it with PARI's implementation of Ryser's formula. It verifies Lemmas 2.1, 2.2 and 4.1 symbolically, and that each of the 120 orderings is the image of exactly one canonical ordering under exactly one affine map (Lemma 4.2). It recomputes the chamber polynomials of Lemma 4.3, not only for the six canonical orderings but for all 120 orderings, which gives $120 \cdot 1001 = 120\,120$ coefficients, of which 119 520 are positive and 600 zero, with the zero coefficients, the least positive coefficient and the four pure-power coefficients as stated. It tests Lemma 2.3 on 200 random rational matrices of orders one to five and identity (3) on 24 random integer pairs; it tests Theorems 3.1

and 3.2 and Corollary 3.3, with their equality cases, on 2000 nonnegative weight vectors, many of them with zero or repeated weights; and it checks the example of Theorem 3.2, the case $n = 2$ of the introduction and the remarks of Section 6. The random tests are evidence, not proofs; the chamber computation, by contrast, is exact and exhaustive.

Checks in this revision. A separate exact Python checker recomputed the order-five defect from the permutation definition of the permanent. It checked all 120 orderings and the unique reduction to six affine representatives. It verified the polynomial identities by inverse binomial transforms. Every chamber has 1001 coefficient positions, with 996 positive coefficients and five zeros. The checker uses the normalization $625P(w)^2 - 24P(g(w))$. Its least positive coefficient is 484. The four pure-gap coefficients agree with Lemma 4.3. This finite check supports the coefficient statistics. It does not replace the Lean proof.

Formal proof in Lean. Lean [4] is a *proof assistant*. It checks a formal proof against its statement using basic logical rules. The results of Section 3 are proved with Lean 4.35.0 and Mathlib [13], its library of formalized mathematics.

The collection library is in `lean/MathAndTheory/Mathematics/GyiresCirculant5`. Its namespace is `MathAndTheory.GyiresCirculant5`. The source is derived from the immutable earlier development [7].

In the library, `circ w` is the circulant with entries `w (j - i)`. The definition `P5` is the polynomial in Lemma 2.2. The definitions `gram` and `conv` give the vectors g and h. The type `Fin 5`, with arithmetic modulo five, represents $\mathbb{Z}/5\mathbb{Z}$.

The principal statements are restated, using only Mathlib's definitions, in a *Challenge* file, which states the theorems with their proofs omitted, the file `lean/Challenges/GyiresCirculant5.lean` (namespace `MathAndTheory.GyiresCirculant5`). There a circulant is a matrix of the form `Matrix.circulant v`, whose entries are `v (i - j)`; this is the transpose of $C(v)$, which describes the same set of matrices. Doubly stochastic matrices are Mathlib's `doublyStochastic`, the permanent is Mathlib's `Matrix.permanent`, $5!/5^5$ is the real quotient `(Nat.factorial 5 : ℝ) / 5 ^ 5`, and J_5 is the constant matrix with entries 5^{-1}. The *Solution* file `lean/Solutions/GyiresCirculant5.lean` proves the same theorems from the library. Lean's Comparator records from the original project report exact agreement with the Challenge statements and acceptance by Lean, NanoDa, and con-ron. Those records are historical evidence. The accompanying verification record gives the current check outcomes separately. The table lists where each statement of the paper is proved; all listed names are in the namespace `MathAndTheory.GyiresCirculant5`.

Statement in this paper	Lean declaration(s)
Theorem 3.1	`strong_inequality`, `strong_inequality_eq_iff`
Theorem 3.2	`strong_margin`, `strong_margin_attained`, `strong_margin_sharp`
Corollary 3.3, homogeneous form	`gyires_homogeneous`, `gyires_homogeneous_eq_iff`
Corollary 3.3, ratio form	`gyires_ratio`, `gyires_ratio_eq_iff`, `gyires_denominator_pos`
Lemma 2.1	`circ_mul_transpose`, `transpose_mul_circ`, `circ_mul_self`
Lemma 2.2	`permanent_circ`
Lemma 2.3, Corollary 2.4	`PermanentGram.permanent_gram_add_ge`, `conv_permanent_le_gram`
Lemma 4.1	`P5_affine`, `gram_affine`, `conv_affine`, `strongDefect_affine`, `gyiresDefect_affine`
Lemma 4.2	`order_affine_cover`, `order_affine_cover_unique`, `canonical_chamber_of_monotone`, `exists_canonical_chamber`
Lemma 4.3, nonnegativity of R_κ	`strong_chamber_0`, ..., `strong_chamber_5`, `strong_chambers`
Theorem 3.2, the example	`strongDefect_twoSupport`, `strongDefect_margin_attained`, `strongDefect_margin_sharp`
Corollary 3.3, positive denominator	`positiveDenominator`
Section 6, the two-point example	`P5_twoSupport`, `P5_gram_twoSupport`, `P5_conv_twoSupport`, `gyiresDefect_twoSupport`, `gyiresDenominator_twoSupport`, `permanent_twoSupport_half`, `permanent_gram_twoSupport_half`, `permanent_square_twoSupport_half`, `normalizedGyiresRatio_twoSupport_half`

The original Lean project records a source build and an axiom audit. The theorem dependencies were listed with Lean's `#print axioms` command. Every one depends only on the three standard axioms of Lean's logic (`propext`, `Classical.choice`, `Quot.sound`); no step is left unproved, and the proofs use neither `native_decide` nor any other evaluation by compiled code. The formal proof of Lemma 4.3 follows the organization described after it: the six chamber polynomials are expanded in z_0 and z_1, and each of the resulting polynomials in z_2, z_3, z_4 is written with explicit nonnegative integer coefficients, proved equal to the corresponding expansion coefficient by the `ring` tactic and nonnegative by the `positivity` tactic. These explicit coefficients were produced by a computer program, but Lean does not rely on that program: it checks every identity and every inequality itself. The formal proof of Lemma 2.3 is not the occupancy argument given here but a sum-of-squares argument over permutations, adapted from the formalization of Ozols [12].

Several remarks are also formalized: the case $n = 2$ of the introduction (`gyires_two`); the characterization of the doubly stochastic circulants of order five and their description as the combinations $\sum_k w_k Q^k$ (`circulant_mem_doublyStochastic_iff`, `circulant_eq_sum_pow`); and, for real square matrices of every order, the reduction of Gyires's inequality to the strong form described in Section 6 (`gyires_of_strong`).

What is not formalized. The formal proof establishes $R_\kappa(z) \geq 0$ for all $z \geq 0$, which is the part of Lemma 4.3 that the theorems need. The finer statements of that lemma (the numbers 996 and 5 of positive and zero coefficients, the positions of the zeros, the four pure-power

coefficients, and the least positive coefficient 484) are not stated in Lean; they are verified by the exact computation described above. The remaining remarks of Section 6, which rest on random tests or on the literature, are not formalized.

6 Discussion

Scope. The results concern circulant matrices of order five only. They do not prove Gyires's conjecture for all of Ω_5, or for any order $n \geq 6$. Within the circulants of order five they cover every matrix in the four-dimensional simplex of doubly stochastic circulants, including all of its faces, and every nonnegative circulant without normalization.

The coordinates. Written in the original weights, D_s has negative coefficients, so its nonnegativity is not visible without the change to sorted gaps. In the chamber coordinates all coefficients are nonnegative, and the zero coefficients are exactly those of the monomials z_0^{10} and $z_0^9 z_j$, which reflects the fact that D_s vanishes to second order along the constant vectors.

The two-point example. For $w = (t, t, 0, 0, 0)$ one has $g(w) = (2t^2, t^2, 0, 0, t^2)$, $h(w) = (t^2, 2t^2, t^2, 0, 0)$, $P(w) = 2t^5$, $P(g(w)) = P(h(w)) = 84t^{10}$, and $D_s(w) = D_G(w) = 484t^{10}$; this is the example showing that 484 cannot be improved. For $t = \frac{1}{2}$ the matrix is doubly stochastic, $\operatorname{per} C = \frac{1}{16}$ and $\operatorname{per}(CC^{\mathsf{T}}) = \operatorname{per}(C^2) = \frac{21}{256}$, and Gyires's ratio equals $\frac{1}{21} \approx 0.048$, above $\frac{24}{625} = 0.0384$. The example also shows that the inequality $\operatorname{per}(AA^{\mathsf{T}}) \leq \operatorname{per}(A)$ fails for some doubly stochastic circulants. If it held, the strong form $\frac{n!}{n^n} \operatorname{per}(AA^{\mathsf{T}}) \leq \operatorname{per}(A)^2$ would follow at once from van der Waerden's bound $\operatorname{per}(A) \geq n!/n^n$. According to Ryser's account, in the translation of Best and Wanless [1], an inequality of this form was once conjectured and was disproved by M. Newman with the order-four analogue $\frac{1}{2}(I + Q)$ of this example, for which $\operatorname{per} A = \frac{1}{8} < \frac{9}{64} = \operatorname{per}(AA^{\mathsf{T}})$.

A stronger question. For every real square matrix A, Lemma 2.3 with $M = A$ and $N = A^{\mathsf{T}}$ gives $2\operatorname{per}(A^2) \leq \operatorname{per}(AA^{\mathsf{T}}) + \operatorname{per}(A^{\mathsf{T}}A)$. Hence the strong form, applied to A and to A^{T}, implies Gyires's inequality for A; this is the reduction used by Ozols [12] for $n = 4$. It is therefore natural to ask whether

$$\frac{n!}{n^n} \operatorname{per}(AA^{\mathsf{T}}) \leq \operatorname{per}(A)^2 \qquad \text{for every } A \in \Omega_n.$$

This holds for $n = 2$ by the computation in the introduction, for $n = 4$ by Ozols's theorem, and for the circulants of order five by Theorem 3.1. The broader statement is not proved here.

Other orders. Every real circulant of order n is normal, and Lemma 2.3 gives $\operatorname{per}(C^2) \leq \operatorname{per}(CC^{\mathsf{T}})$ in every order, so Gyires's conjecture for circulants of order n follows from the strong form $\frac{n!}{n^n} \operatorname{per}(CC^{\mathsf{T}}) \leq \operatorname{per}(C)^2$. This paper makes no claim for higher orders. The method of this paper depends essentially on the value of n: the affine group of $\mathbb{Z}/n\mathbb{Z}$ reduces the $n!$ orderings of the weights to $(n-2)!$ chambers when n is prime, and the degree of the defect grows with n.

About this paper

Contributions and review. Dev D. Goyal selected the problem, requested proofs and checks, and accepts responsibility for publication. He did not supply proof ideas or intermediate arguments. The source repository [7] records that ChatGPT (OpenAI), using the GPT-6

model with Pro reasoning, constructed the proof and original Lean development. It records that Claude Opus 5.5, used with Claude Code, revised the formalization and manuscript and performed additional checks.

Codex, using GPT-6, audited the mathematical arguments and source declarations, reconstructed the exact coefficient certificate independently, and prepared the present revision. The collection records the separate AI referee review and the completed verification checks. The recorded personal review by Dev D. Goyal applies to the earlier proof, not to the new formal statements or this revised manuscript. Machine verification and AI review do not constitute independent human expert review. The paper has not been peer reviewed.

Prior-work limits. The audit consulted the sources listed here and used exact-name and equivalent-term web searches. It inspected the existing order-four and order-five repositories and the current Lavi catalogue. The original project also records earlier searches of several Lean collections. Those earlier searches are historical evidence, not fresh searches by this revision. Gyires's original 1978 article and the full Chang article were not retrieved. The author-hosted Cheon–Wanless PDF returned an access error, although its bibliographic record and relevant indexed text were available. The live Lavi catalogue was retrieved on 26 October 2026. MathSciNet, zbMATH, and complete Google Scholar citation chaining were not completed. These limits prevent an unconditional claim of priority.

Source record. The existing order-five repository is cited at an immutable commit [7]. This revision preserves the old repository and its contribution notices. The collection is at `https://github.com/dvgyl/math-and-theory`. Its verification report identifies the files and checks used for the current revision.

Acknowledgements

The conjecture is due to B. Gyires. The formal proof of Lemma 2.3 is adapted from the Lean formalization of the case $n = 4$ by C. Ozols [12], which is licensed under the Apache License 2.0. Thanks to the developers of Lean and Mathlib.

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